Innovation Centre for Education

Low-Level Design

for

Hardened Cloud Infrastructure for Retail Data

Mohammed Saqib

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BCA in Cloud Computing, Cyber Security & Ethical Hacking with IBM

Yenepoya Institute of Arts, Science, Commerce and Management

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Under the guidance of

Ms. Munisha

Lecturer, Computer Science Department
Yenepoya Institute of Arts, Science, Commerce and Management

Mangalore

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Table of Content

1. Introduction
   1.1 Scope of the Document
   1.2 Intended Audience
   1.3 System Overview
2. System Design
   2.1 Application Architecture Diagram
   2.2 Process Flow (Dispatch Lifecycle) – Sequence Diagram
   2.3 Information Flow (Telemetry Engine) – Data Flow Diagram
   2.4 Components Design (Detailed)
   2.5 Key Design Considerations
   2.6 API Catalogue
3. Data Design
   3.1 Entity-Relationship (ER) Diagram
   3.2 Data Model (Schema & Encryption)
   3.3 Data Access Mechanism
   3.4 Data Retention & Archival Policies
   3.5 Data Migration & Seeding
4. Interfaces
   4.1 User Interface (UI) Layout
   4.2 API Contracts (Request/Response Structure)
5. State and Session Management
6. Caching Strategy
7. Non-Functional Requirements
   7.1 Security Aspects
   7.2 Performance Aspects
8. Conclusion

1. Introduction

1.1 Scope of the Document

This Low-Level Design (LLD) document provides detailed technical specifications for implementing the Layered Retail Cloud Security framework. It covers Azure infrastructure provisioning (VNet, subnets, NSG, Application Gateway with WAF_v2), the Python-based encryption engine (Argon2id key derivation, AES-256-GCM authenticated encryption), Flask API endpoints for secure file upload/download, database schema for metadata and audit logs, session management, caching strategy, and non-functional requirements (security + performance). The document serves as the complete implementation blueprint for DevOps engineers and Python developers.

1.2 Intended Audience

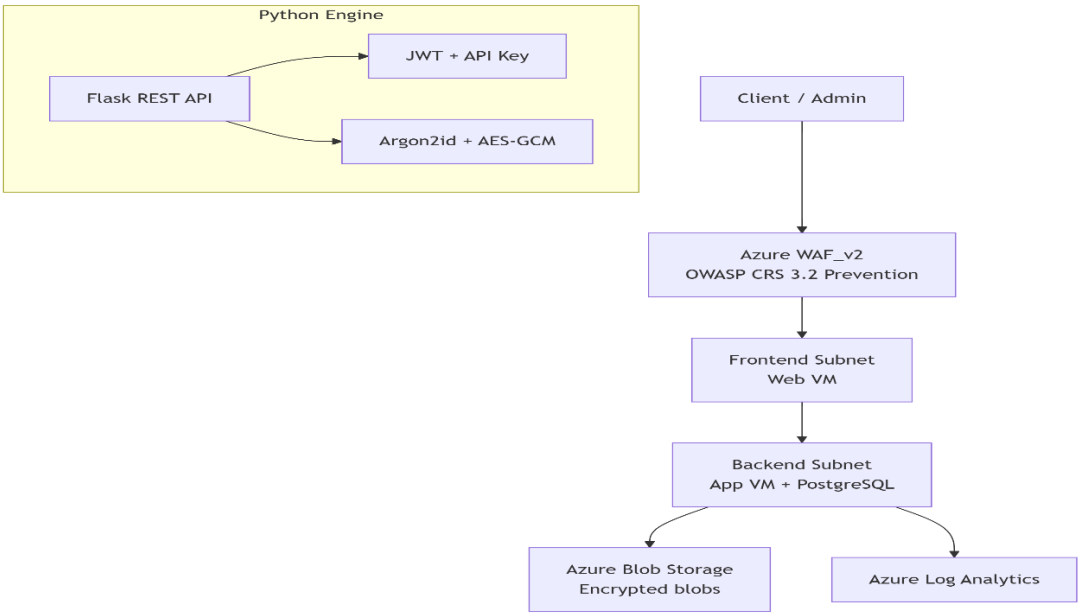
- **Cloud Engineers –** Provision Azure resources using Terraform.
- **Developers –** Implement the Python encryption engine and REST API.
- **Security Testers –** Validate WAF rules, NSG isolation, and cryptographic correctness.
- **Retail IT Administrators –** Understand the defense-in-depth controls.

1.3 System Overview

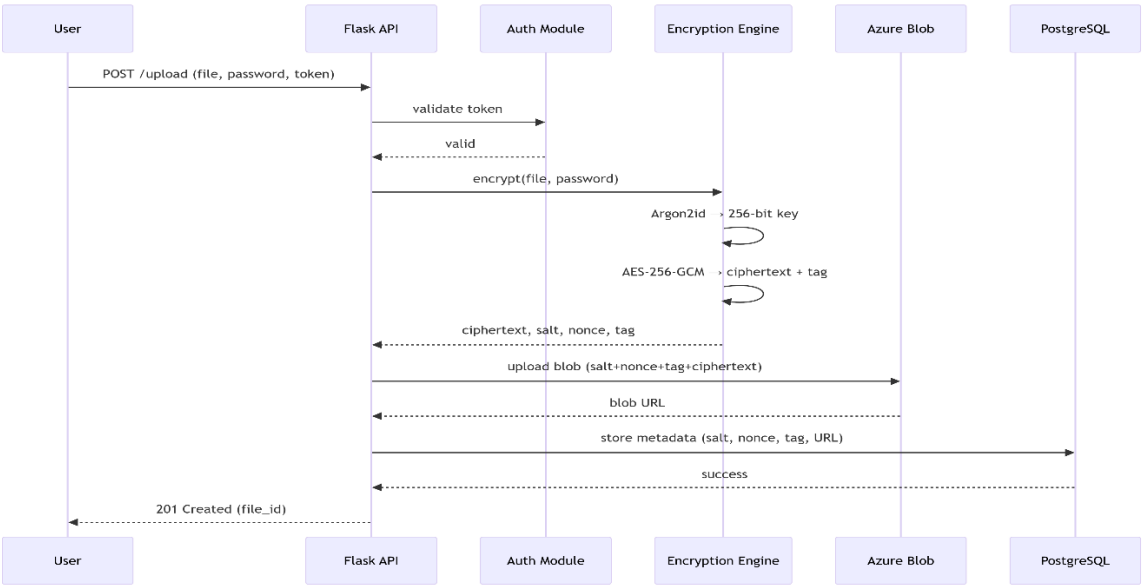
Flask-based web application on Microsoft Azure that implements defense-in-depth for retail data. Azure WAF_v2 (OWASP CRS 3.2) blocks SQLi/XSS at the perimeter. A VNet with frontend and backend subnets isolates the application VM and database; NSG rules allow backend access only from the frontend. Data-in-transit uses TLS 1.3. For data-at-rest, a Python engine derives a 256-bit key via Argon2id (64MB memory, 3 iterations) and encrypts files using AES-256-GCM. The ciphertext (with salt, nonce, tag) goes to Azure Blob Storage; metadata to PostgreSQL. Access is controlled via stateless JWT tokens with RBAC (Admin/User/Auditor). All events log to Azure Log Analytics. Even if the perimeter is breached, retail data stays encrypted.

2. System Design

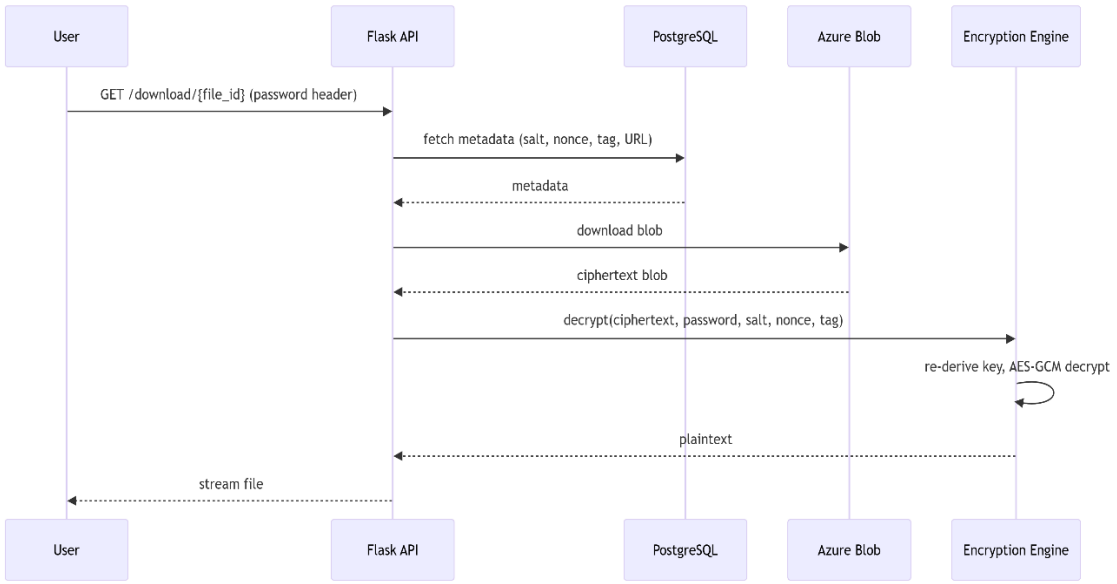
2.1 Application Design



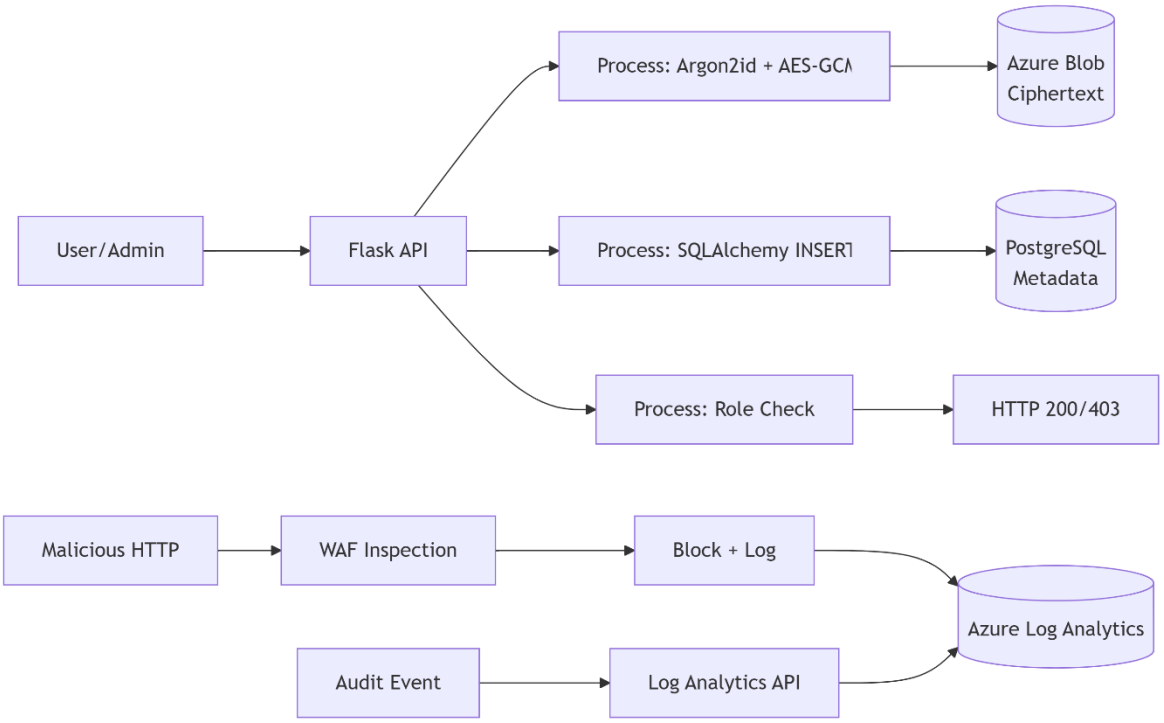
2.2 Process Flow (Dispatch Lifecycle) – Sequence Diagram:
Upload (Encryption) Sequence:



Download (Decryption) Sequence:



2.3 Information Flow (Telemetry Engine) – Data Flow Diagram



2.4 Components Design

Component	Technology	Specifications
Key Derivation	Argon2id (`argon2-cffi`)	Memory=64 MB, Time=3, Parallelism=1, Salt=16 B
Symmetric Cipher	AES-256-GCM (`cryptography`)	Nonce=12 B, Tag=16 B
Binary File Structure	Custom	`[Salt(16)][Nonce(12)][Tag(16)][Ciphertext]`
Infrastructure	Terraform 1.4+, Azure CLI	VNet `10.0.0.0/16`, subnets: frontend `10.0.1.0/24`, backend `10.0.2.0/24`
NSG Rule (Backend)	Azure NSG	Allow TCP 80,443 from `10.0.1.0/24` only
WAF	Azure WAF_v2	OWASP CRS 3.2, Prevention mode, log all triggers

2.5 Key Design Considerations

- **Defense-in-Depth – No single layer is trusted; overlapping controls.**
- **Zero-Trust Data – Encrypted before storage; cloud provider never sees plaintext.**
- **Performance – Argon2id memory cost fits within 2 GB VM.**
- **Compliance – AES-256-GCM meets FIPS 140-2 for data-at-rest.**

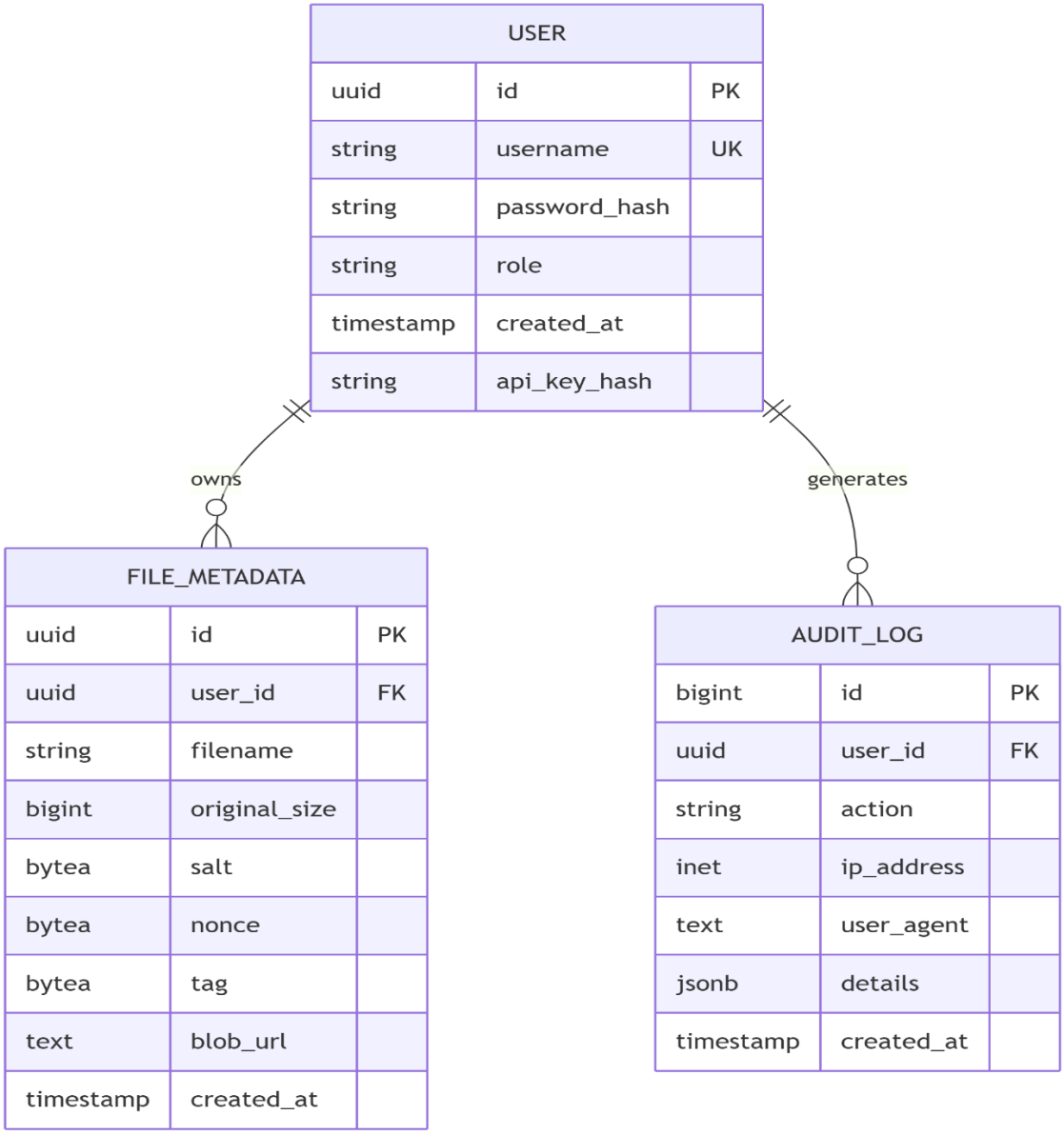
2.6 API Catalogue

Endpoint	Method	Description	Auth
`/upload`	POST	Upload and encrypt a file	JWT or API Key
`/download/{file_id}`	GET	Download and decrypt a file	JWT or API Key + password header
`/health`	GET	Health check	None

`/audit`	GET	Retrieve audit logs (admin only)	JWT (admin role)
`/login`	POST	Obtain JWT token	None

3. Data Design

3.1 Entity-Relationship (ER) Diagram



3.2 Data Model (Schema & Encryption)

- **Table users –** stores authentication data (passwords hashed with PBKDF2-SHA256).
- **Table file_metadata –** stores encryption parameters (salt, nonce, tag) and blob URL. Actual ciphertext is in Azure Blob.
- **Table audit_log –** logs all security events.
- **No plaintext or encryption keys are ever stored** – keys are derived ephemerally from user password

3.3 Data Access Mechanism

- **ORM:** SQLAlchemy with connection pooling.
- **Queries:** User.query.filter_by(username=username).first()
- **Transactions:** db.session.add() and commit().
- **Blob operations:** Azure Blob SDK upload_blob / download_blob.

3.4 Data Retention & Archival Policies

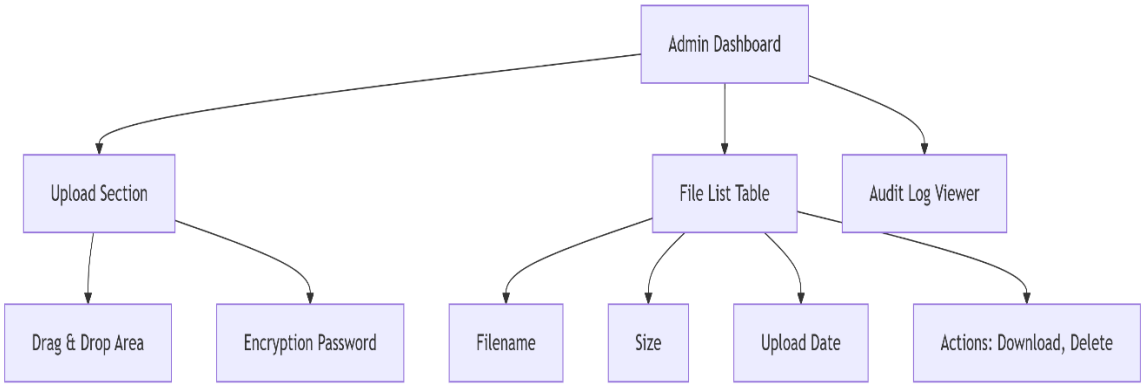
- **User accounts**: Permanent until deletion request.
- **File metadata & blobs:** Deleted when user account is removed.
- **Audit logs:** 180 days retention (Azure Log Analytics).
- **Archival:** Files >90 days old moved to Azure Cool Blob tier.

3.5 Data Migration & Seeding

- **Migration tool:** Alembic (Flask-Migrate).
- **Seed data:** 100 sample retail files pre-encrypted for testing.
- **Rollback:** Database snapshots + Alembic downgrade.

4. Interfaces

4.1 User Interface (UI) Layout



4.2 API Contracts (WebSocket Events)

POST /upload

Item	Description
Request Format	multipart/form-data
Request Fields	- file: binary file content - password: user-supplied string for key derivation
Response (Success – 201 Created)	file_id (UUID) and message ("File encrypted and stored")
Response (Failure)	Appropriate HTTP error code (400, 401, 413, etc.)

GET /download/{file_id}

Item	Description
Request Headers	`X-User-Password`: password string for decryption
Response (Success – 200 OK)	Binary file stream (original plaintext file)
Response (401)	Wrong password
Response (404)	File not found

POST /login

Item	Description
Request Body	`username` and `password` (plain text)
Response (200 OK)	`access_token` (JWT, expires 15 min), `refresh_token` (cookie)
Response (401)	Invalid credentials

GET /audit (admin only)

Item	Description
Request Header	`Authorization: Bearer <admin_jwt>`
Response (200 OK)	JSON array of audit log entries (action, timestamp, user, ip, details)
Response (403)	Insufficient role (non-admin)

5. State and Session Management

- **Authentication:** Stateless JWT (15 min expiry) + refresh token (7 days, HTTP-only cookie).
- **No server-side session store –** horizontal scaling ready.
- **Logout:** Client discards token; refresh cookie cleared.
- **Secure flags:** Refresh token cookie HttpOnly, Secure, SameSite=Strict.

6. Caching

- **Redis (**Azure Cache for Redis)
- **Cached item:** Derived encryption key derived_key:{hash(password+salt)} – TTL 5 minutes.
- **Cache invalidation:** On password change, delete all keys for that user.
- **No caching of plaintext –** security requirement.
- **Static assets**: Browser caching only.

7. Non-Functional Requirements

7.1 Security Aspects

Aspect	Implementation
Password storage (API auth)	PBKDF2-SHA256 (Werkzeug)
Data-at-rest encryption	AES-256-GCM + Argon2id
Data-in-transit	TLS 1.3 (Azure Gateway)
WAF	OWASP CRS 3.2 Prevention
Network isolation	NSG allow only frontend subnet
SQL injection	SQLAlchemy ORM
XSS	Jinja2 auto-escape (if UI added)
CSRF	Stateless JWT avoids CSRF
Audit logging	Azure Log Analytics

7.2 Performance Aspects

Metric	Target	Means
Encryption throughput	≥ 50 MB/s	AES-NI hardware acceleration
Argon2id latency	≤ 300 ms	64 MB memory, 3 iterations
API response (p95)	≤ 500 ms (excl. upload)	Redis cache, connection pooling
Concurrent users	100	Horizontal scaling
Uptime	99.9%	Azure availability zones + health probes

Conclusion

The Layered Retail Cloud Security LLD provides a complete technical blueprint for a defense-in-depth system. It combines Azure network security (WAF, NSG, VNet isolation) with a strong cryptographic engine (Argon2id + AES-256-GCM). The design ensures that even if the network perimeter is breached, retail data remains encrypted and inaccessible. All components are developer-ready, scalable, and compliant with modern security standards.